

SQL Practice Set — GROUP BY, HAVING & LEFT JOIN

Dataset: Urban Bike-Sharing Network | 8 Questions | Dialect: MySQL 8.0

Read the schema and sample data below carefully, then write one SQL query per question so that your result matches the expected output exactly — same columns, same column names, same row order.

Schema

```
CREATE TABLE stations (  
  station_id INT PRIMARY KEY,  
  station_name VARCHAR(50),  
  zone VARCHAR(20),  
  capacity INT,  
  opened_year INT  
);  
  
CREATE TABLE rides (  
  ride_id INT PRIMARY KEY,  
  station_id INT,  
  rider_type VARCHAR(20),  
  ride_date DATE,  
  duration_min INT,  
  distance_km DECIMAL(4,1),  
  fare DECIMAL(6,2),  
  FOREIGN KEY (station_id) REFERENCES stations(station_id)  
);
```

Sample Data

Table 1: stations (9 rows)

station_id	station_name	zone	capacity	opened_year
1	Riverside Dock	North	20	2019
2	Amber Plaza	Central	35	2018
3	Kite Hill	North	15	2021
4	Old Mill Yard	East	25	2017
5	Glass Bridge	Central	40	2020
6	Fern Gully	West	12	2022
7	Copper Lane	East	18	2016
8	Willow Bend	West	22	2023
9	Stone Arch	South	30	2023

Table 2: rides (25 rows)

ride_id	station_id	rider_type	ride_date	duration_min	distance_km	fare
1	1	Member	2024-03-02	22	4.2	3.50
2	1	Casual	2024-03-05	45	7.8	9.75
3	2	Member	2024-03-05	18	3.1	2.80
4	2	Student	2024-03-11	33	6.0	4.20
5	3	Casual	2024-03-14	12	1.9	2.10
6	1	Member	2024-04-01	27	5.4	4.10
7	4	Student	2024-04-03	55	9.5	6.30
8	5	Member	2024-04-07	15	2.6	2.40
9	5	Casual	2024-04-09	62	11.2	13.50
10	2	Member	2024-04-12	24	4.8	3.90
11	6	Student	2024-04-18	40	7.1	5.00
12	3	Member	2024-05-02	19	3.3	2.95
13	7	Casual	2024-05-04	50	8.9	11.00
14	5	Student	2024-05-08	29	5.2	3.75
15	1	Casual	2024-05-13	36	6.4	8.20
16	4	Member	2024-05-15	21	4.0	3.30
17	2	Casual	2024-05-19	48	8.1	10.40
18	6	Member	2024-05-22	17	2.9	2.55
19	5	Member	2024-06-01	31	6.7	4.60
20	7	Student	2024-06-04	44	7.6	5.40
21	1	Student	2024-06-09	26	4.9	3.60
22	3	Casual	2024-06-11	58	10.3	12.25
23	4	Casual	2024-06-15	39	6.8	8.60
24	2	Member	2024-06-20	23	4.4	3.70
25	5	Casual	2024-06-25	53	9.7	12.80

Questions

Question 1

From the rides table, show each rider_type along with the total number of rides taken and the total fare collected. Include only those rider types that have taken more than 7 rides. Name the output columns rider_type, total_rides and total_fare, and sort the result by total_fare in descending order.

Expected Output:

riders_type	total_rides	total_fare
Casual	9	88.60
Member	10	33.80

Question 2

For every station_id in the rides table, display the number of rides and the average ride duration rounded to 2 decimal places. Show only those stations whose average duration is greater than 30 minutes. Name the output columns station_id, ride_count and avg_duration, and sort the result by avg_duration in descending order.

Expected Output:

station_id	ride_count	avg_duration
7	2	47.00
4	3	38.33
5	5	38.00
1	5	31.20

Question 3

Group the rides by the month of ride_date and display the month number, the number of rides in that month, and the total distance covered rounded to 1 decimal place. Show only the months where the total distance is greater than 30 km. Name the output columns ride_month, total_rides and total_distance, and sort by ride_month in ascending order.

Expected Output:

ride_month	total_rides	total_distance
4	6	40.6
5	7	38.8
6	7	50.4

Question 4

For each rider_type, display the longest ride duration, the shortest ride duration and the average ride duration rounded to 2 decimal places. Include only those rider types whose longest ride is 50 minutes or more. Name the output columns rider_type, longest_ride, shortest_ride and avg_ride, and sort by longest_ride in descending order.

Expected Output:

rider_type	longest_ride	shortest_ride	avg_ride
Casual	62	12	44.78
Student	55	26	37.83

Question 5

Group the rides by station_id and rider_type together. For each combination, display the number of rides and the total distance rounded to 1 decimal place. Show only the combinations where the total distance is greater than 10 km. Name the output columns station_id, rider_type, ride_count and total_distance, and sort by station_id in ascending order.

Expected Output:

station_id	rider_type	ride_count	total_distance
1	Casual	2	14.2
2	Member	3	12.3
3	Casual	2	12.2
5	Casual	2	20.9

Question 6

Find the stations that have been used by all three rider types (Member, Casual and Student). For each such station display the number of distinct rider types, the total number of rides and the average fare rounded to 2 decimal places. Name the output columns station_id, rider_type_count, total_rides and avg_fare, and sort by station_id in ascending order.

Expected Output:

station_id	rider_type_count	total_rides	avg_fare
1	3	5	5.83
2	3	5	5.00
4	3	3	6.07
5	3	5	7.41

Question 7

List every station in the stations table along with the details of its June 2024 rides — ride_id, rider_type and fare. Stations that had no ride in June 2024 must still appear in the result with NULL values in the ride columns. Name the output columns station_id, station_name, zone, ride_id, rider_type and fare, and sort by station_id and then ride_id.

Expected Output:

station_id	station_name	zone	ride_id	rider_type	fare
1	Riverside Dock	North	21	Student	3.60
2	Amber Plaza	Central	24	Member	3.70
3	Kite Hill	North	22	Casual	12.25
4	Old Mill Yard	East	23	Casual	8.60
5	Glass Bridge	Central	19	Member	4.60
5	Glass Bridge	Central	25	Casual	12.80
6	Fern Gully	West	NULL	NULL	NULL

7	Copper Lane	East	20	Student	5.40
8	Willow Bend	West	NULL	NULL	NULL
9	Stone Arch	South	NULL	NULL	NULL

Question 8

Produce a station-wise summary showing station_id, station_name, zone, the total number of rides started at that station and the total revenue collected. Every station must appear in the result — stations with no rides should show 0 rides and 0.00 revenue instead of NULL. Name the output columns station_id, station_name, zone, total_rides and total_revenue, and sort by total_rides in descending order, then by station_id in ascending order.

Expected Output:

station_id	station_name	zone	total_rides	total_revenue
1	Riverside Dock	North	5	29.15
2	Amber Plaza	Central	5	25.00
5	Glass Bridge	Central	5	37.05
3	Kite Hill	North	3	17.30
4	Old Mill Yard	East	3	18.20
6	Fern Gully	West	2	7.55
7	Copper Lane	East	2	16.40
8	Willow Bend	West	0	0.00
9	Stone Arch	South	0	0.00

Question 9

Explain what are the types of SQL languages

Question 10

Explain what are joins in SQL and different types of joins with Venn diagram